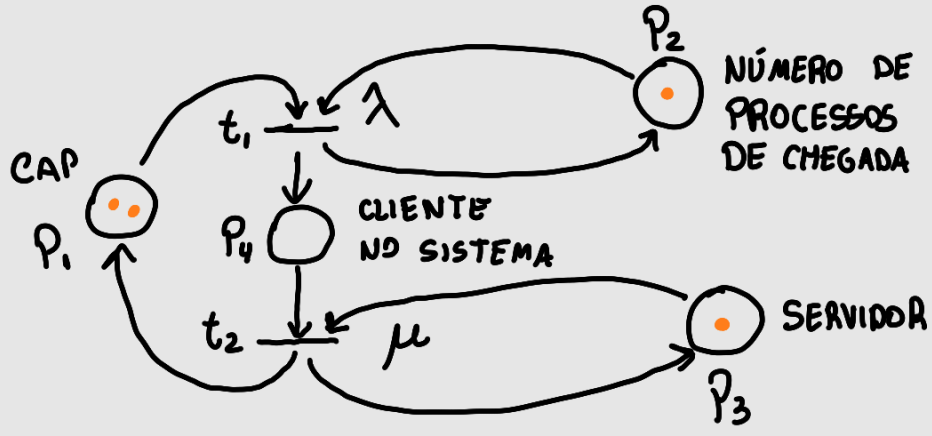
① M/M/1/2

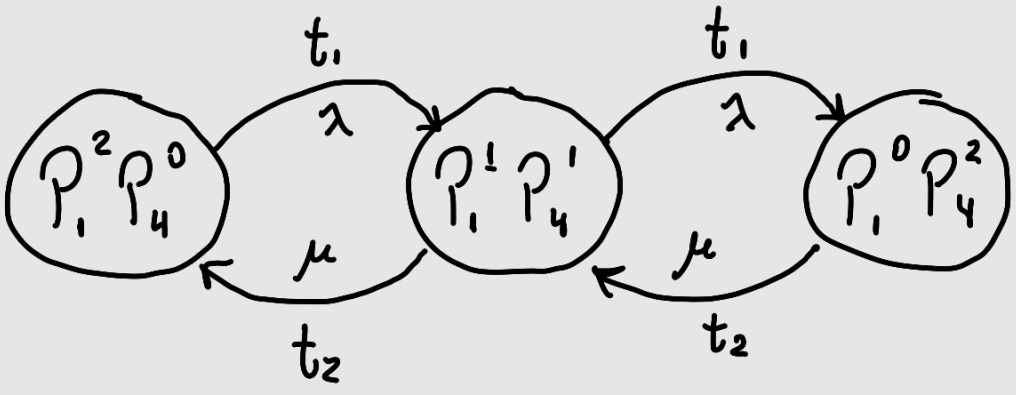
$\lambda = 1\,min^{-1}$

$\frac{1}{\mu} : 1\,min \rightarrow \mu = 1\,min^{-1}$

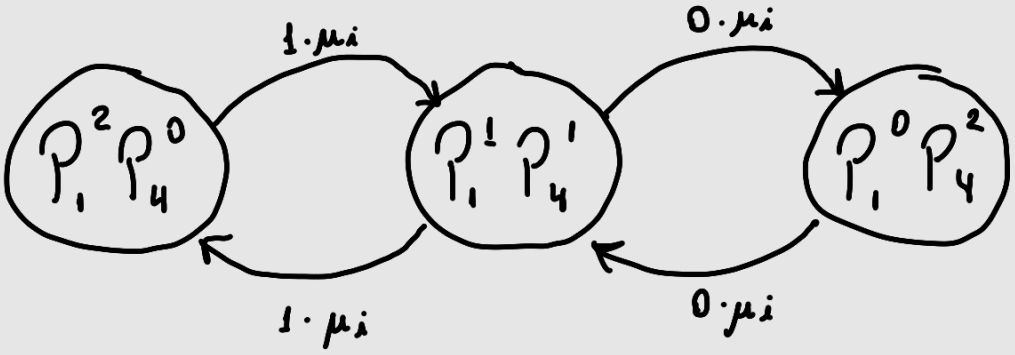
a)



b)

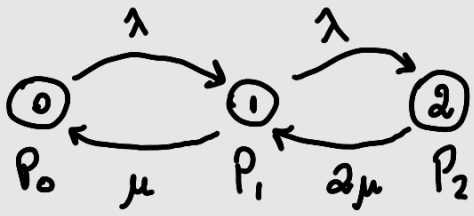


c)



d)

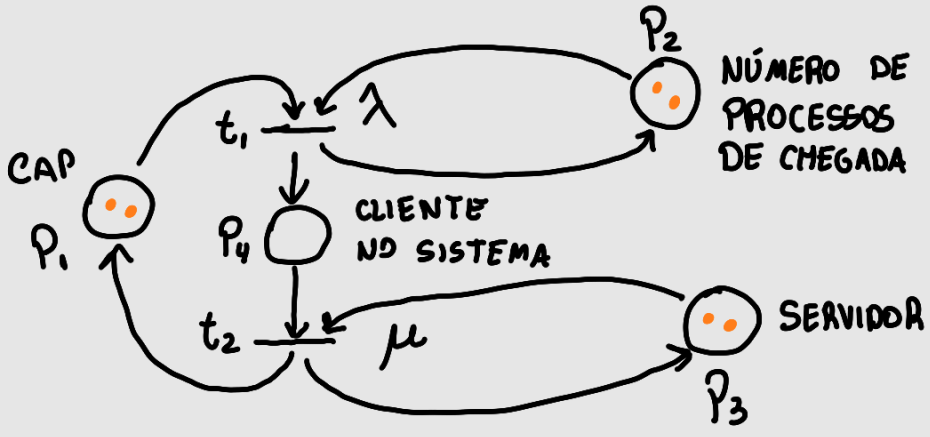
② M|M|2|2



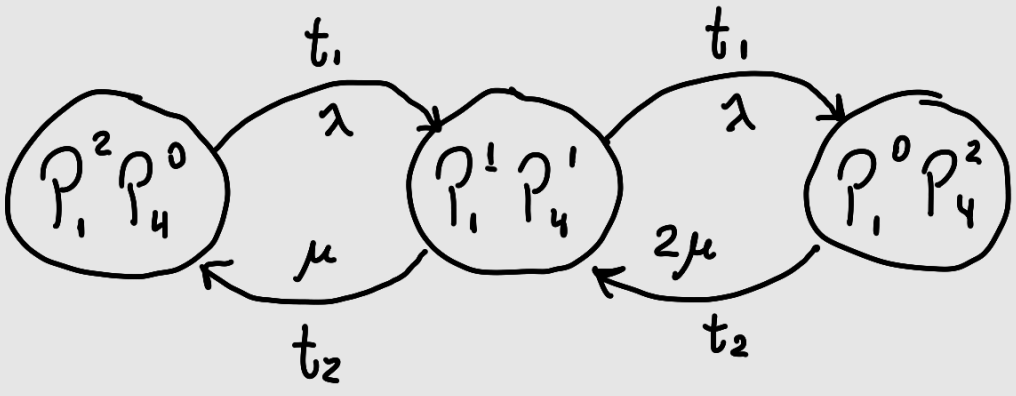
$\lambda = 1\ min^{-1}$

$\frac{1}{\mu} = 1\ min \rightarrow \mu = 1\ min^{-1}$

a)



b)



c)





d) Considerando $\lambda = \mu = 1\,min^{-1}$

Est. 0 : $\mu \cdot P_1 = \lambda \cdot P_0$

Est. 1 : $\lambda P_0 + 2\mu \cdot P_2 = \mu \cdot P_1 + \lambda \cdot P_1$

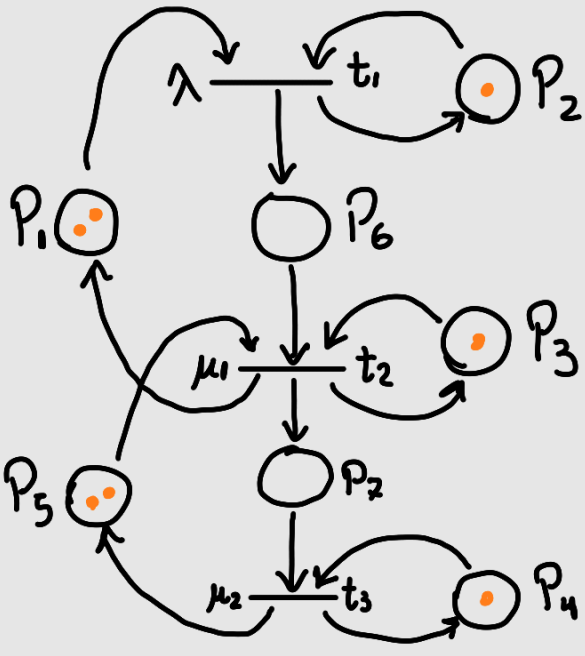
Est. 2 : $\lambda \cdot P_1 = 2\mu \cdot P_2$

Norm. : $P_0 + P_1 + P_2 = 1$

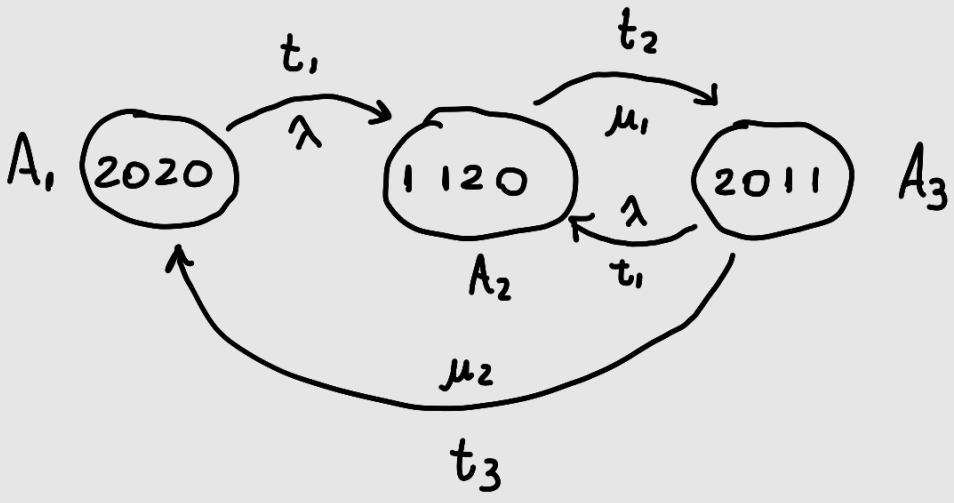
$P_0 = \frac{2}{5}$ $\quad$ $P_1 = \frac{2}{5}$ $\quad$ $P_2 = \frac{1}{5}$

∴ A porcentagem de clientes atendidos será $P_0 + P_1 = 80\%$

3



a) Os estados serão representados pela quantidade de fichas em P_1, P_6, P_5 e P_7, respectivamente.



b)

